Multi-regional machine learning for adolescent mental health: a synthesis of three studies with PeNSE 2019

Thalles Paul Leandro Mota

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Abstract. *This report brings together three studies published in 2026 that apply machine learning to Brazil's National School Health Survey (PeNSE 2019) to identify adolescents at risk, always evaluating the models in states that took no part in training. The first classified the risk of frequent substance use in the Center-West, validated in Mato Grosso (AUROC 0.89; recall 0.80). The second classified psychological distress nationwide, training on one state per macro-region and evaluating on the remaining 22 (AUROC 0.87; F2 0.71), and showed that sensitivity holds without the sadness variable. The third asked where the difficulty of these problems comes from and found that it depends more on how the target is defined than on the variables available. Together, the studies form a protocol of geographic validation, stable interpretation and diagnosis of limits that applies to screening in public health.*

1. Why I studied this

Adolescent mental health is a social problem, and Brazil has laws to protect young people. Researchers around the world now use machine learning to flag adolescents at risk, but few look hard at whether the variables a model relies on are stable, whether it can be trusted outside the data it was trained on, and how its findings relate to what epidemiology already knows. That is what I set out to examine during my master's degree, and these three studies are the result.

2. Context

Adolescence concentrates vulnerabilities that range from substance use to suicidal behavior; suicide is the second leading cause of death among young people aged 15 to 24 worldwide [Gerstner et al. 2018]. In Brazil, PeNSE, run by the national statistics institute (IBGE), collects health data from school students across the country [IBGE 2021], which makes it possible to build classifiers for screening. A country of continental size, however, imposes a demand that studies with random validation do not test: a model built with data from some states has to work in the others.

The three studies, carried out with C. A. Martins, L. C. O. Gonçalves, A. C. S. de Oliveira and L. A. V. Garcia, answer complementary questions: (i) can risk be classified well in states the model has not seen? (ii) are the criteria the models learn stable and interpretable? (iii) where does the performance ceiling come from?

3. Common design

The studies share the data source (PeNSE 2019), an emphasis on sensitivity, since in screening missing an adolescent at risk costs more than a false alarm, and geographic

validation: training uses some states, and evaluation uses states the model never saw. Table 1 summarizes each study.

Table 1. The three studies.

	Study 1 [Mota and Martins 2026]	Study 2 [Mota et al. 2026a]	Study 3 [Mota et al. 2026b]
Outcome	frequent substance use (3 or more days out of 30)	psychological distress (feeling that life is not worth living most of the time or always)	frequent substance use, in three variants
Scope	Center-West	Brazil	Brazil
Geographic validation	trained on MS, GO and DF; tested on MT	trained on one state per macro-region; tested on the remaining 22 states; 10 draws	same design as study 2; 10 draws
Models	XGBoost and random forest; Optuna (300 iterations)	XGBoost, CatBoost and LightGBM; Optuna (12,000 iterations)	committee of 4 models and a control neural network
Central question	is risk detectable outside the training states?	are the criteria stable and auditable?	does the difficulty come from the variables or from the target?

4. Study 1: frequent substance use in the Center-West

Trained on Mato Grosso do Sul, Goiás and the Federal District (10,348 records, 5.3% prevalence) and validated only on Mato Grosso (3,277 records, 5.6% prevalence), XGBoost tuned for recall reached, across the ten reported configurations, an AUROC of 0.89, recall between 0.80 and 0.81 and precision of about 0.23 [Mota and Martins 2026]. XGBoost kept recall more stable than the random forest. The SHAP analysis pointed to having friends who smoke as the strongest predictor, followed by recent alcohol use and being sexually active, and identified protective factors such as family supervision and parental support.

5. Study 2: psychological distress nationwide

With 105,116 valid records and a prevalence of 19.8% (from 17.3% in the South to 22.0% in the North), the study drew one state per macro-region for training and evaluated the models on the remaining 22, repeating the draw ten times [Mota et al. 2026a]. To separate real criteria from artifacts of the search, five stability indices were defined over the SHAP values of the best-performing models, and these criteria were compared with each variable's statistical association with the outcome (Cramér's V). The correlation was positive in every draw, and a core of variables recurred in all of them, including body self-perception, parental support, bullying and attending a public school.

In unseen states, XGBoost reached an AUROC of 0.87, F2 of 0.71 and recall of 0.84. Without the sadness variable, the one most associated with the outcome, AUROC and F2 fell to 0.80 and 0.64, but recall barely changed (0.83; no significant difference in the paired Wilcoxon test): screening ability holds with sociodemographic and context variables. An additive risk index built directly from the SHAP weights (RII), with no

modeling of interactions, came within 0.016–0.021 F2 of the classifier, which makes it an auditable instrument. Performance varied between macro-regions in a stable way, with the Center-West at the top and the Northeast at the bottom (Figure 1).

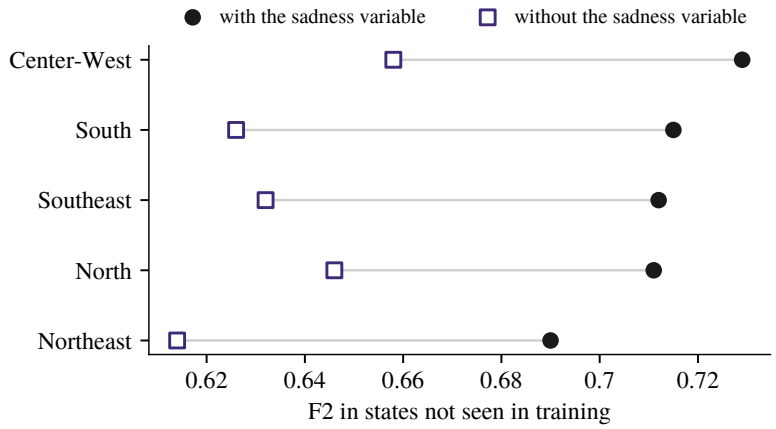


Figure 1. Median F2 of the combined system (RII and classifier) by macro-region, in states outside training, with and without the sadness variable (ten draws; data from Table 5 of [Mota et al. 2026a]).

6. Study 3: where the difficulty comes from

With 165,838 adolescents and a prevalence of about 4%, the third study measured how hard each instance is in two independent ways: by the error propensity of a committee of four models (instance hardness) and by the local overlap between classes (kDN), which uses no model [Mota et al. 2026b, Smith et al. 2014]. Difficulty turned out to be largely a property of the data: agreement between the four models exceeded 0.94 (Spearman). Local overlap explained only part of the errors (AUC of about 0.73, against 0.95 for the empirical measure), which points to label noise or non-local interactions.

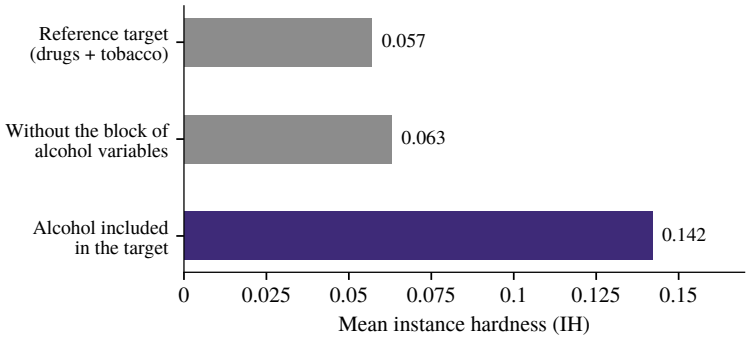


Figure 2. Mean instance hardness in three variants of the same problem (ten draws; data from [Mota et al. 2026b]).

The central result is in Figure 2: removing the whole block of alcohol variables, keeping the same instances and the same target, barely changed the difficulty (0.057 to 0.063; median paired difference of -0.0004), while widening the target to include alcohol raised it about 2.5 times (0.057 to 0.142), with the committee's error rate rising from 9%

to 18%. In this domain, once a reasonable set of variables is in place, the performance ceiling depends more on what is being predicted than on the variables used to predict it.

7. What the three studies teach together

1. **Validating outside the training data changes the conclusion.** Evaluation in unseen states reveals regional differences that a random split would hide; thresholds calibrated by region absorb part of them.
2. **In screening, the metric chooses the model.** Prioritizing recall (or F2) accepts more false alarms in exchange for not missing adolescents at risk; that choice has to be made explicitly.
3. **An interpretation is only worth something if it is stable.** Criteria that recur across draws and agree with an independent statistical association can become auditable instruments, such as the RII.
4. **Before looking for more variables, look at the target.** The definition of the outcome can be the main source of difficulty.

The same logic (train on part of the data sources, evaluate on the others and diagnose where the errors come from) applies to any model that has to work outside the setting in which it was built.

8. Limitations

The data are from 2019, before the pandemic. The outcomes are self-reported and work as indicators of distress or risk, not as a clinical diagnosis; the systems were designed for screening and monitoring. In study 2, the survey's sampling weights were not used, because the aim is individual prediction rather than population inference, and the blocks of the regional analysis are not fully independent. In the same study, a fairness check by skin color or race showed recall between 0.81 and 0.87 across the five groups, and a broader analysis is left for future work [Mota et al. 2026a].

References

Gerstner, R. M. F., Soriano, I., Sanhueza, A., Caffe, S. and Kestel, D. (2018). Epidemiología del suicidio en adolescentes y jóvenes en Ecuador. *Revista Panamericana de Salud Pública*, 42:1–7.

IBGE (2021). Pesquisa Nacional de Saúde do Escolar: 2019. Rio de Janeiro: IBGE.

Mota, T. P. L. and Martins, C. A. (2026). Aprendizado de máquina na classificação do risco de uso frequente de substâncias em adolescentes na região Centro-Oeste. In *Anais do Congresso de Inteligência Artificial de Mato Grosso*. doi:10.29327/9786527222910.1448987.

Mota, T. P. L., Gonçalves, L. C. O., Oliveira, A. C. S. and Martins, C. A. (2026). Classifying psychological distress in Brazilian adolescents: a multi-regional machine learning approach. In *Intelligent Systems (BRACIS 2026)*, LNAI 17109, p. 546–560. Springer. doi:10.1007/978-3-032-39904-5_37.

Mota, T., Garcia, L. A. V. and Martins, C. (2026). Where does hardness come from? An instance hardness analysis of substance-use prediction in Brazilian adolescents. In *Anais do XIV KDMiLe*, p. 265–272. SBC. doi:10.5753/kdmile.2026.31988.

Smith, M. R., Martinez, T. and Giraud-Carrier, C. (2014). An instance level analysis of data complexity. *Machine Learning*, 95:225–256.